Every Fraction of a Degree Counts for the Planet

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Countries are gathered in Turkey for COP31, the climate talks, to tackle a goal that is slipping away. Ten years ago, nations agreed to limit warming to 1.5 degrees Celsius above **pre-industrial** levels. Now, scientists say that goal is out of reach.

The planet is nearly 1.5 degrees warmer than it was before the Industrial Revolution. Heat-trapping emissions from burning **fossil fuels** are still rising, and most countries are not on track to meet their climate promises to switch to cleaner energy.

A United Nations study says the best case now is 1.7 or 1.8 degrees of warming. If emissions stay on their current path, the world could be 2.8 degrees hotter by the end of the century.

The difference between 1.5 and 2 degrees may sound tiny, but research shows it changes everything. "The crucial point is that every fraction of a degree matters," says Tim Lenton, a climate professor at the University of Exeter. "Every fraction of a degree that we avoid is really able to do a huge amount of limiting harm."

Coral reefs show why. Corals build the framework that supports vast numbers of other species. But corals are very sensitive to heat. Long marine heat waves turn them white, a process called **bleaching**, which can eventually kill them. There have been four global **bleaching** events since the late 1990s, and scientists say a fifth may now be unfolding.

Researchers found that before the 2023 to 2025 **bleaching** event, 44 percent of coral species were at risk of **extinction**. Beyond 2 degrees of warming, the Intergovernmental Panel on Climate Change found, 99 percent of corals would be lost.

Countries are focusing on **electrification**: cars and homes running on electricity instead of **fossil fuels**, since much of the world's electricity now comes from solar and wind.

Word Treasure

pre-industrial -- The time before factories and machines started changing the climate, long ago.

fossil fuels -- Fuels like coal, oil, and gas that release heat-trapping gases when burned.

bleaching -- When hot water makes corals turn white and lose the tiny living things inside them.

extinction -- When a kind of living thing dies out completely and is gone forever.

electrification -- Switching cars and homes to run on electricity instead of fuels that pollute.

Background you need

At climate talks like COP31, countries from around the world meet to agree on how to limit global warming and cut the heat-trapping gases released when fossil fuels are burned.

The goal they set was to keep warming to 1.5 degrees Celsius above the temperatures before the Industrial Revolution, when factories and machines began burning large amounts of coal, oil, and gas.

Even small amounts of extra warming matter because hotter oceans stress coral reefs, which provide homes and shelter for many other sea creatures, and long heat waves can turn corals white and kill them.

One way countries are trying to cut emissions is electrification, which means powering cars and homes with electricity, much of which now comes from cleaner sources like solar and wind.

Break it down

LEAD: At COP31 in Turkey, countries are tackling a 1.5-degree warming limit that scientists now say is out of reach.

KEY EVENT: The planet is nearly 1.5 degrees warmer than before the Industrial Revolution, while heat-trapping emissions from burning fossil fuels keep rising.

+- WHO: Countries gathered at COP31, scientists, and a United Nations study

+- WHAT: Warming may reach 1.7 or 1.8 degrees in the best case, or 2.8 degrees by the end of the century on the current path

+- WHEN: Ten years after nations agreed the 1.5-degree goal; the 2023 to 2025 bleaching event is the latest warning

EVIDENCE: Before the 2023 to 2025 bleaching event, 44 percent of coral species were at risk of extinction, and beyond 2 degrees of warming, 99 percent of corals would be lost, according to the Intergovernmental Panel on Climate Change.

RESPONSE: Countries are focusing on electrification, with cars and homes running on electricity from solar and wind instead of fossil fuels.

OPEN QUESTION: How much warming can still be avoided, since every fraction of a degree matters?

Why it matters

Every fraction of a degree of warming decides how much of the natural world survives, from coral reefs to the weather children will grow up with, so the choices made at COP31 shape the planet they will inherit.

Test what you remember

1. **Where are countries gathered for the COP31 climate talks?**
   - (A) Turkey
   - (B) France
   - (C) Egypt
   - (D) Brazil
2. **What warming limit did nations agree to ten years ago?**
   - (A) 1.7 degrees Celsius above pre-industrial levels
   - (B) 2.8 degrees Celsius above pre-industrial levels
   - (C) 2 degrees Celsius above pre-industrial levels
   - (D) 1.5 degrees Celsius above pre-industrial levels
3. **According to a United Nations study, what is the best case for warming now?**
   - (A) 2.0 degrees
   - (B) 2.8 degrees
   - (C) 1.5 degrees
   - (D) 1.7 or 1.8 degrees
4. **If emissions stay on their current path, how much hotter could the world be by the end of the century?**
   - (A) 1.8 degrees
   - (B) 3.5 degrees
   - (C) 2.8 degrees
   - (D) 1.5 degrees
5. **What happens to corals during long marine heat waves?**
   - (A) They grow faster and spread to new areas
   - (B) They produce more food for other species
   - (C) They turn white in a process called bleaching, which can eventually kill them
   - (D) They move to colder water
6. **Before the 2023 to 2025 bleaching event, what share of coral species were at risk of extinction?**
   - (A) 44 percent
   - (B) 15 percent
   - (C) 99 percent
   - (D) 25 percent

Answer key (try the quiz first!): 1: A 2: D 3: D 4: C 5: C 6: A

Questions to think about

1. Tim Lenton, a climate professor at the University of Exeter: Every fraction of a degree matters, and every fraction of a degree that we avoid is really able to do a huge amount of limiting harm.

2. The Intergovernmental Panel on Climate Change: Beyond 2 degrees of warming, 99 percent of corals would be lost.

MY THOUGHTS (at least 20 words):